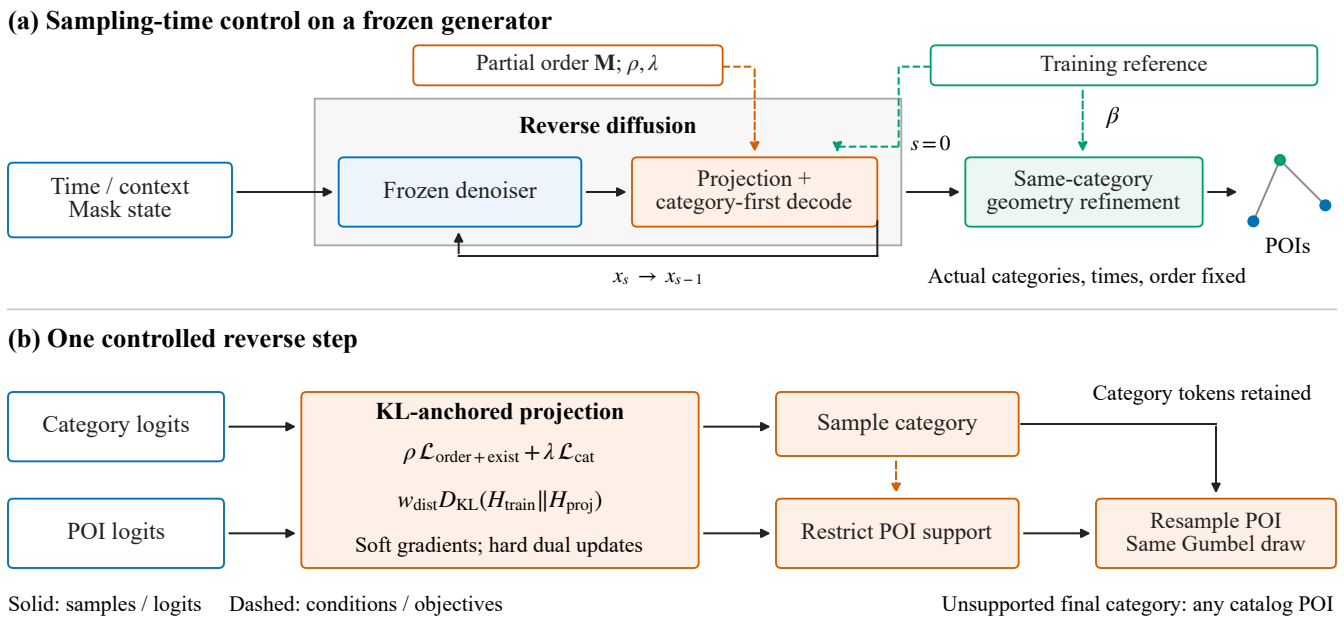

(a) Sampling-time control on a frozen generator
Partial order M; ρ, λ
Training reference
Reverse diffusion
β
s = 0
Time / context
Mask state
Frozen denoiser
Projection +
category-first decode
Same-category
geometry refinement
POIs
x_s → x_{s−1}
Actual categories, times, order fixed
(b) One controlled reverse step
Category logits
POI logits
KL-anchored projection
ρ L_{order+exist} + λ L_{cat}
w_{dist} D_{KL}(H_{train} ‖ H_{proj})
Soft gradients; hard dual updates
Sample category
Category tokens retained
Restrict POI support
Resample POI
Same Gumbel draw
Solid: samples / logits
Dashed: conditions / objectives
Unsupported final category: any catalog POI